Technische Universität München · TUM School of Management

Green Steel from Western Australia

A Trade-Model Analysis of Thyssenkrupp's Investment and Export Strategy

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Professorship of Economics of Energy Markets
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1 Introduction

Thyssenkrupp Steel Europe (TKSE) is Germany's largest flat-steel producer. Using iron ore imported to its integrated Duisburg site, the company manufactures hot-rolled coil (HRC). Naturally, this operation is subject to the high cost of European energy. This paper will examine whether TKSE should build a fully integrated, green HRC plant in Western Australia and ship output to the European market. HRC is a suitable unit of analysis because it is a commodity-grade, internationally traded intermediate whose landed cost governs competitiveness across suppliers.

The EU is among the world's largest steel markets, with total consumption of roughly 134 Mt in 2025 and flat-steel demand near 76 Mt, of which over 40% is imported (Eurofer, 2026). Competition in HRC is essentially based on cost. In early 2026, standard north-west European HRC traded around €710/t ex-works (Fastmarkets, 2026b), while certified green HRC carried a premium of roughly €100/t, trading at about €800/t (Fastmarkets, 2026a). A small set of European producers are beginning to increase certified low-carbon production. This means that the green segment trades at a premium but remains thinly supplied relative to projected automotive demand for low-CO_2 steel, which is projected to be at 31 Mt by 2035 (Discovery Alert, 2026). Automakers are among the largest consumers of flat steel and have some of the strongest decarbonization commitments, which makes them the segment most willing to pay for certified green coil (Fastmarkets, 2026c). 2025 saw a softening in green premiums and thin trading volumes, showing that the premiums are not guaranteed.

Western Australia is attractive to TKSE because it removes the three cost disadvantages of the Duisburg base. The Pilbara region, the planned location for the integrated plant, already holds the iron ore, meaning that none would have to be imported (Geoscience Australia, 2025; U.S. Geological Survey, 2025). It also has among the world's highest solar and wind potential, which will deliver cheap renewable power (Devlin et al., 2023). In turn, this will provide TKSE with cheap green hydrogen that direct reduction iron, an intermediary good, requires for production (Devlin et al., 2023). In short, the ore and the clean energy that Duisburg must source separately and at high costs are already accumulated in one place.

Moreover, Western Australia's location advantage is reinforced by industry and policy. The Green Iron Investment Fund backs domestic green-iron investment (Department of Industry, Science and Resources, 2025) and POSCO's roughly 3bn USD green-iron project, located at Port Hedland, provides a real-world benchmark for a plant of this scale (GMK Center, 2026). Recent modeling also identifies Australia as one of the most competitive locations for green H_2-DRI steel serving the European market from 2026 onward (Johnson et al., 2025). On the policy side, the EU's Carbon Border Adjustment Mechanism (CBAM), introduced under Regulation 2023/956, phases in from 2.5% of the full carbon charge in 2026 to 100% by 2034 (Regulation (EU) 2023/956). At a certificate price of about €75/t CO_2 (CBAM Guide, 2026) and a carbon intensity of about 2.7 t CO_2e/t for Chinese blast-furnace HRC (Hasanbeigi, 2025), CBAM raises the EU-delivered price of carbon-intensive imports far more than that of certified green steel. Thus, it widens the delivered-cost gap in TKSE's favor and strengthens the case for producing green steel in Australia rather than continuing to produce conventional steel in Germany.

The central decision taken by TKSE is best examined as three interrelated questions, each addressed through an established model of international trade. Markusen's theory of the multinational firm will evaluate the location and structure of production best suited for TKSE (Markusen, 2002). The pricing of a differentiated variety will be looked at through Krugman's model of monopolistic competition (Krugman, 1980). Finally, the firm's viability will be calculated using Melitz's model of firm heterogeneity (Melitz, 2003). Together, the three models will answer the following questions.

First, given Western Australia's combined advantages in iron ore, renewable energy, and green hydrogen, should TKSE relocate integrated green-HRC production abroad and serve the EU through exports rather than continue producing in Germany? Second, does TKSE's certified green HRC constitute a differentiated variety rather than commodity coil? If so, does pricing it at the corresponding elasticity of substitution sustain a markup that covers the plant's cost, particularly as CBAM raises the delivered prices of its carbon-intensive rivals? Third, is the Australian plant productive and cost-efficient enough to survive? Secondary questions this paper will examine using the Melitz model are: does TKSE's production cost fall below the market's threshold cost, and does the margin widen as CBAM phases in toward 2034?

The three models are applied in sequence in the next section. I will synthesize the output of each model in the final section as well as consider the limitations of the analysis and implications for the market, the industry, and trade policy.

2 Methodology

The three models are applied in sequence, each resolving the question previously left open. Markusen (2002) decides where TKSE should locate production. Krugman (1980) decides how the resulting variety should be priced. Melitz (2003) decides whether the plant is efficient enough to survive at that price. The three models sit together rather than compete, and consequently, they draw on a single shared parameter base (Table 1). This allows for figures established in one section to carry into the next without restatement, creating consistency. Two symbols recur across the three models with meanings defined by each model's standard usage. The subscript i indexes the home country, the EU, in the Markusen and Melitz sections, and a single variety in the CES pricing block. The starred quantity, q*, indicates each model's own optimal output which are the Krugman break-even quantity in pricing and the plant capacity in the Markusen calibration.

Table 1: *Shared parameter set for the three-model analysis*

Symbol	Description	Value
α	Demand intercept (maximum willingness to pay)	€1,040/t
β	Demand slope	€95/t Mt
L_i	EU market size (normalized)	1.0
L_j	Australia market size (relative to EU)	0.1425
MC_i	German green H_2-DRI-EAF production cost	€800/t
$MC_{j,\,local}$	Australian production cost	€510/t
MC_j	Australian coil delivered to the EU (= 510 + τ)	€565/t
τ	Freight, Australia → EU	€55/t
G_i	German plant fixed cost	€329M/yr
G_j	Australian plant fixed cost	€439M/yr
F	Firm-specific fixed cost	€100M/yr

Symbol	Description	Value
w	Australian wage	€96,000/yr

Note. All costs are stated in € per ton (€/t) or € million per year (€M/yr). Demand parameters (α, β) are calibrated from the 2025 EU HRC market equilibrium; cost and fixed-cost parameters are drawn from case-study estimates and the sources compiled in the reference list.

2.1 Markusen: On the Location Decision

Assuming a monopolistically competitive firm, Markusen's (2002) framework evaluates the location choice by comparing organizational structures. The framework hinges on four assumptions. First, there are two country markets, which in our case study are Germany/the EU (i, TKSE's home and primary market) and Western Australia (j, the host). The relative market sizes are $L_i = 1.0$ and $L_j = 0.1425$. Second, the existence of a single differentiated variety of green HRC sold under linear inverse demand, $p = \alpha - \beta X$, in each. The existence of plant-specific fixed costs, G, alongside a firm-specific fixed cost, F, with cross-border freight, τ, of €55/t. Fourth, a German benchmark MC_i of €800/t is used, reflecting the green H_2 DRI–EAF route in Duisburg rather than the incumbent BF-BOF process so that the exercise can isolate location rather than technology. This €800/t German cost is close to the current market price of certified green HRC. Producing green steel in Germany earns almost no margin, which is the core reason the Australian route, at €510/t, is worth evaluating. Furthermore, no subsidies are credited which returns conservative output and additionally biases the resulting comparison against relocation.

Three organizational structures are evaluated. Following the domestic structure, (Type-d), TKSE produces only in Germany and exports to serve the Australian market. Under the horizontal structure, (Type-h), TKSE operates a plant in each country. Finally, the vertical-FDI structure, (Type-v), positions TKSE to produce only in Australia and ship finished coil back to the EU. For a monopolist facing linear demand, profit in each market, m, takes the form,

$$\Pi_m = (\alpha - MC_m)^2 \cdot L_m / (4\beta) \quad (1)$$

Total profit sums the two market contributions net of the fixed costs of each structure, with $4\beta = 380$.

Table 2 shows the output when evaluating Equation 1 across the three structures. The domestic structure serves the EU from Germany at €800/t, and Australia at €855/t landed cost; thus, contributing €151.6M and €12.8M respectively. When calculating net of G_i and F, the structure loses €264.6M per year. The horizontal structure substitutes the Australian export with local production at €510/t, benchmarked off the Port Hedland project, raising the Australian contribution to 105.3. However, the second plant adds additional annualized plant cost, G_j, and produces the largest loss of €611.1M per year. The vertical structure serves the EU from Australia at a landed cost of €565/t. Through contributions of €593.8M and €105.3M locally, the structure returns a profit of €160.1M per year inclusive of annualized plant cost for Western Australia and fixed overhead costs.

Table 2: *Annual profit by organizational structure*

Structure	EU segment	AUS segment	Fixed costs	Total Π
Type-d (domestic)	151.6	12.8	$G_i + F = 429$	−264.6
Type-h (horizontal)	151.6	105.3	$G_i + G_j + F = 868$	−611.1
Type-v (vertical FDI)	**593.8**	**105.3**	**$G_j + F = 539$**	**+160.1**

Note: Values in €M/yr, computed from Equation 1 (4β = 380) on the Table 1 parameters. The EU and AUS columns are per-market profit contributions before fixed costs.

As demonstrated above, Vertical FDI is the only profitable structure, by an extensive margin. Relocating to Australia lowers the marginal cost of serving the large EU market by €235/t, shown by €800 compared to €565 delivered. Producing solely abroad also allows TKSE to avoid the German plant cost, Gi, entirely. Holding the remaining parameters fixed, the vertical structure stays profitable up to a plant fixed cost of approximately €599M per year. Crediting the larger subsidies available to a German plant would narrow the lead but not overturn it. The €160.1M/yr result answers where TKSE should produce but takes the sales price as whatever the linear demand monopoly equation returns. Whether certified green HRC is differentiated enough to command a markup is the question that Krugman will examine, while using the same €510/t marginal cost as a pricing base.

2.2 Krugman: Differentiation and the Price of the Variety

The Markusen model established that TKSE should produce in Western Australia and export to the EU, under the vertical FDI optimal structure. It assumed a monopoly and took the sales price as the outcome of a linear demand schedule. The sustainability of said price depends on something the Markusen model does not capture. The question Krugman asks is whether TKSE's certified green HRC is a differentiated variety that buyers will pay a premium for; and if so, what price can it sustain. Krugman's (1980) model of monopolistic competition supplies the mechanism. It rests on three assumptions. First, firms operate under increasing returns to scale, so each variety is made by a single plant holding high fixed costs. Second, consumers display love for variety, valuing differentiated products under constant elasticity of substitution (CES) demand. The strength of that preference is governed by a single parameter, which is elasticity (ε). Third, free entry drives profit to zero in the long run. For calculations using the Krugman model, the EU is the sales market. Market size (wL), the price index (P), and the number of varieties are EU variables, while production sits in Australia at the marginal cost derived from Markusen at €510/t.

Under CES demand, a firm supplying variety i charges a constant markup over marginal cost,

$$p_i = (\varepsilon/(\varepsilon-1)) \cdot (w/\varphi_i) = (\varepsilon/(\varepsilon-1)) \cdot MC \quad (2)$$

where the marginal cost is written as the wage over labor productivity, w/φ_i. Free entry then fixes the zero-profit, or minimum efficient, quantity and the number of varieties the market supports,

$$q_i = (\varepsilon-1) \cdot f / MC \quad (3)$$

$$n^* = wL / (\varepsilon \cdot f) \quad (4)$$

with $f = G_j + F$ = €539M representing the annual fixed cost carried over from the Markusen model. Everything in this section turns on a single parameter, ε. The constant elasticity of substitution (CES) generates both the markup in Equation 2 and determines whether green HRC can command a premium at all.

The elasticity: a single, argued value.

In CES, ε represents how readily buyers switch between varieties and, therefore, the model's measure of switching costs and customer loyalty. A high ε describes near-interchangeable

goods, buyers who take on no cost in changing away from a supplier, and almost zero pricing power or, in other words, a thin markup. Conversely, a low ε describes a differentiated product to which buyers are attached and face real switching costs, meaning that the seller holds real pricing power.

There is no measured elasticity for green HRC. To address this, ε is placed between two anchors that are measured, both from the Harmonized-System trade data. When evaluated purely as a commodity, Bajzik et al.'s (2020) analysis puts ε for steel near 3.8. This is consistent with Broda and Weinstein's (2006) finding that primary metals sit at the high, interchangeable end. Calibrating ε from current prices would give that same commodity value, since today's green and gray coil still trade almost interchangeably. At the differentiated end of the spectrum, the average macro elasticity, defined in Feenstra et al. (2018) as the substitutability between home and imported goods, is around 2. Thus, the placement follows a known pattern. The USITC survey conducted by Ahmad, Montgomery, and Schreiber (2021) finds that commodities sit high on the scale while differentiated goods sit low. Consequently, as certification makes green HRC less interchangeable, its elasticity should move down toward the differentiated anchor.

Moreover, two further considerations ground defensibility that ε can be assumed to be 2. First, the type of elasticity evaluated is consistent with Feenstra et al. (2018). The relevant substitution for TKSE is between EU-domestic certified green coil and imported commodity coil. In other words, a home versus import margin. Second, ε at 2 can also be considered a willingness-to-pay floor. Below this elasticity, the CES markup implies prices that no buyer would accept. The value is carried as a single constant with a transparent 1.8 – 2.5 sensitivity band.

It is held fixed, deliberately. The switching costs rising as certification spreads and locks buyers in would require a year-by-year path, which no published estimate supports. Instead, the CBAM and Scope 3 schedules are what show the change over time. The advantage widens because policy raises competitors' costs on a fixed legal timeline, not because ε is assumed to move. Growth in switching costs over the period is left out of the model entirely, which further pushes the conservative nature of the evaluation.

The value of differentiation

Pricing the two regimes through the markup rule, in Equation 2, quantifies what the differentiation is worth. If treated as a commodity, ε equaling 3.8, TKSE's variety would sell at €692/t. As this is below the €710/t benchmark, there is no premium and no case can be made for carrying the fixed cost. However, when treated as a differentiated variety, with ε equal to 2, the coil would sell at €1,020/t. The €328/t gap between the two regime prices is the economic value of differentiation, as shown in Figure 1 and Table 3. The investment case rests on establishing the fact that green HRC belongs with the second regime.

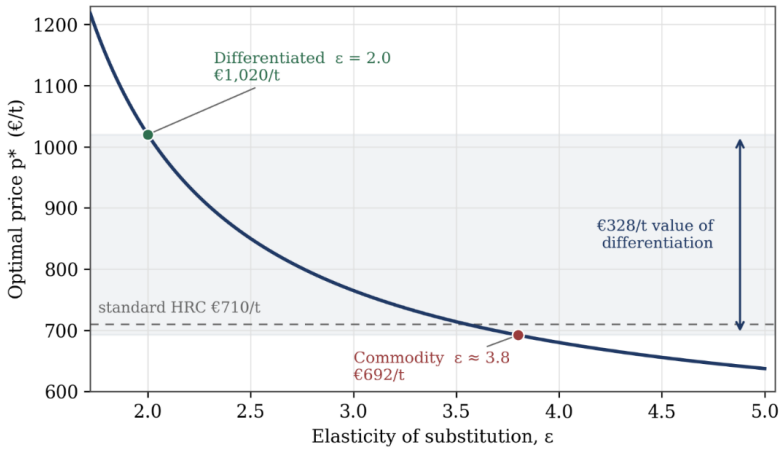
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Table 3: *The value of differentiation: two elasticity regimes*

Regime	ε (source)	markup	p*	premium vs €710	break-even q*	viability
Commodity coil	3.8 (Bajzik et al., 2020)	1.36	€692/t	−€18/t	2.96 Mt	below standard HRC; no premium — unviable
Differentiated variety	**2.0 (Feenstra et al., 2018)**	**2.00**	**€1,020/t**	**+€310/t**	**1.06 Mt**	**healthy markup; clears break-even with room**

Note. Prices from Equation 2 at MC = €510/t. Premium is measured against the €710/t NW-European standard-HRC benchmark. The €328/t gap (€1,020 − €692) is the economic value of differentiation. The €710/t benchmark and the demand intercept draw on the same Fastmarkets Northern Europe HRC series at slightly different dates in early 2026, consequently the small drift between them does not affect any result.

Figure 1: *Optimal price as a function of the elasticity of substitution*



Note. p* = [ε/(ε−1)] · €510. The curve steepens sharply as ε falls: moving green HRC from the commodity anchor (ε ≈ 3.8) to the differentiated anchor (ε = 2.0) raises the sustainable price by €328/t.

In the differentiated regime, the base case gives a price (p*) of €1,020/t, a break-even quantity (q*) of 1.06 Mt/y, and an equilibrium variety count (n*) of roughly 23. The break-even quantity sits significantly below the planned quantity of 2.5 Mt/y, so viability holds with a wide margin. It also survives the sensitivity band comparison. When ε is equal to 2.5, which is comparable to partial differentiation, the price is €850/t and the break-even quantity rises, slightly, to 1.59 Mt/y. Note that ε equal to 1.8 is excluded as it falls below the willingness to pay floor, as seen in Table 4.

Table 4: *Base case and sensitivity band (differentiated regime, f = €539M)*

ε	markup	p*	premium vs €710	q* (Mt)	n*	note
2.5	1.667	€850	+€140	1.59	18.6	conservative — partial differentiation
2.3	1.769	€902	+€192	1.37	20.2	mid-band
2.0	**2.000**	**€1,020**	**+€310**	**1.06**	**23.2**	**base case (Feenstra macro; WTP floor)**

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ε	markup	p*	premium vs €710	q* (Mt)	n*	note
1.8	2.250	€1,148	+€438	0.85	25.8	below WTP floor — excluded

Note. q* from Equation 3, n* from Equation 4 with wL ≈ €25bn (EU HRC market). The base case (ε = 2.0) is the Feenstra macro elasticity and the WTP floor; ε = 1.8 is reported only to mark the lower bound.

The base-case premium is €310/t over the €710/t standard, above the €100–200/t that differential green flat-rolled coil trades at today (Fastmarkets, 2026a). That is high for the broad market. The plant, though, is not built for the broad market. It is built for automakers, who already sign long-term contracts for certified low-carbon coil at €200–350/t (Kotas, 2026). The €310/t premium falls inside that band, so it is a price the target segment already pays. Broader buyer surveys are more cautious. Across all steel users, only about one in five pay more than €70/t, and almost none above €305/t (Fastmarkets, 2024). That figure is a floor for the market as a whole, not the ceiling for the automotive contracts that matter here. These figures will be realized in the 2030s, once CBAM has raised the gray baseline and Scope 3 accounting, which counts the carbon embedded in purchased steel toward a automaker's own emissions target, has made green non-optional for automotive buyers.

2.3 Melitz: Firm Heterogeneity and Survival

Krugman established that, when priced as a differentiated variety, TKSE's green HRC sustains p* equal to €1,020/t on an Australian marginal cost of €510/t. Additionally, the break-even point of roughly 1.06 Mt/yr sits comfortably inside what is planned for the plant. This settles how the variety is priced, but not whether the plant is efficient enough to earn that price without loss. Melitz (2003) shows that, in a market of heterogeneous producers, a firm operates only if its productivity clears a certain threshold. Equivalently, from the cost side, the firm must operate at a marginal cost that is below a break-even ceiling. Therefore, the Melitz question is whether the productivity that it implies, through φ = w/MC, clears the threshold set by the model and whether the Australian plant's €510/t cost sits far enough below that ceiling to survive.

Calculations continue directly on the Markusen profit formulation from Section 2.1. The winning vertical – FDI structure survives as long as its annual profit is nonnegative, $\Pi^v \geq 0$. The break-even cost, c*, is the marginal cost at which the profit falls to zero, $\Pi^v(c^*) = 0$. Melitz re-expresses that cost as productivity through MC = w/φ; marginal cost is the wage divided by output per worker. This allows the cost ceiling to map directly onto a productivity floor,

$$\varphi^* = w / c^* \tag{5}$$

The break-even cost, c*, is obtained by setting the vertical-FDI profit to zero. Substituting the vertical structure into the profit function (Equation 1), with the Australian plant producing at cost, c, and delivering to the EU at c + τ, yields a quadratic. The unknown in this equation is the gap between the market's willingness to pay and the break-even cost. The two market sizes, the freight charges, and the combined fixed cost are all coefficients that are fixed entirely by case parameters. Two roots exist; however, one is discarded as it describes a plant that could never price its output at a profit. The admissible root gives a cost gap of €471.16, and a break-even cost, c*, of €568.84/t. Substituting this figure back into the original profit function returns a profit of €0, confirming the solution.

Expressed as productivity and evaluated at the Australian annual wage of €96,000, the plant reaches 188.2 t/emp/yr, clearing the threshold of 168.8 t/emp/yr by a factor of 1.12 (Table 5).

Looking at the cost element, the same result is a 10% cost cushion of €58.84/t between the plant's €510 cost and the €568.84 break-even ceiling. It is important to remember that the wage does not drive this outcome. Wage enters the plant's productivity and the threshold simultaneously, meaning that it cancels by comparison. The outcome, productivity above the threshold and cost below the ceiling, holds regardless of the precise wage.

Table 5: *Melitz survival test in productivity and cost forms*

Measure	Value	vs threshold	Verdict
Model productivity φ = w/MC	**188.2 t/emp/yr**	**1.12× over**	**clears the bar**
Threshold productivity φ*	168.8 t/emp/yr	—	minimum to survive
TKSE Australian cost (MC)	€510/t	€58.84/t under	clears on cost
Threshold cost c*	€568.84/t	—	maximum cost to survive

Note. φ = w/MC and φ* = w/c* (Equation 5) at w = €96,000/yr (ABS mining AWOTE); c* from the break-even condition Π^v = 0 (Equation 1). The productivity and cost rows are the same test in two units.

Serving two markets rather than one lowers the threshold. If TKSE were to supply in the EU, the productivity threshold would rise to 180.3 t/emp/yr, from 168.8 t/emp/yr. The break-even cost would fall to roughly €532/t, from €568.84/t. The Australian market, with no local green-HRC producers, absorbs part of the fixed cost and relaxes the threshold. This idea of local sales buffering EU demand swings and widening the margin of survival is the Melitz "entry with export" logic at work.

That said, the static cushion understates the case. Three concepts move in TKSE's favor over the lifetime of the plant. Firstly, the Australian marginal cost is assumed to fall as time goes on. According to PwC Australia's trajectory, green-hydrogen costs decline from 5.70 AUD/kg in 2025 toward 2.00 AUD/kg by 2040. This is important because the plant uses, roughly, 55kg of hydrogen per ton of DRI. Every 1AUD reduction in the cost of hydrogen means a decrease in the marginal cost of green-HRC of 55AUD/t. Secondly, CBAM raises rivals' delivered costs on its legislated schedule. This will lift the effective demand intercept and the break-even ceiling, c*. Finally, the hardest aspect for the model to quantify is the effect of experience. As the plant matures, operational learning lowers costs and industry-respected certifications, such as blue-mint®, defend the plant's price premium as competitors try to catch up.

Below, Table 6 shows the combined effect. As CBAM begins to take effect, the break even ceiling, c*, rises and the required productivity, φ*, falls. The cost cushion widens from about €59/t in 2026 to €240/t by 2034, with the steepest percent increase in the 2030 phase-in. Consequently, the plant is the most valuable if it is operational before 2030.

Table 6: *Cost cushion widening through 2034 (ε held constant)*

Year	MC (€/t)	α_eff (€/t)	c* (€/t)	φ* (t/emp/yr)	Cushion (c*−MC)
2026 (base)	510	1,040	568.8	168.8	€59/t (—)
2028	480	1,045	573.8	167.3	€94/t (1.6×)
2030 (CBAM phase-in)	430	1,065	593.8	161.7	€164/t (2.8×)
2034	**380**	**1,091**	**619.8**	**154.9**	**€240/t (4.1×)**

Note. c* = α_eff − 471.16, the break-even from Equation 1; φ* = w/c* at w = €96,000/yr. α_eff rises by about one-third of the rival CBAM add-on, which runs from +€3.8/t in 2026 to +€153.8/t in 2034; MC_j follows the PwC Australia green-H_2 trajectory. Cushion = c* − MC.

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Two points summarize the mechanism and close the analysis. First, the widening is driven jointly by marginal cost falling due to the decrease in price of hydrogen used in intermediary goods and the rising break-even productivity threshold as CBAM is implemented. The wage cancels throughout the calculations, so the trajectory is a comment about productivity units, production costs, and the carbon-border schedule, not about the wage assumption. Second, CBAM enters the survival analysis on the cost side. It raises rivals' delivered costs, which lifts the break-even ceiling c* and widens TKSE's cushion over time, without touching TKSE's own price or markup.

3 Discussion and Conclusions

3.1 Synthesis

The recommendation to build in Western Australia does not rest on the output of any single model, but rather the point at which all three converge. Markusen, Krugman, and Melitz each answer a different question. Where to locate, how to price, and whether the plant is efficient enough to survive. Each returns the same answer, in different contexts, placing the strength of the case in their agreement. What makes the case worth making is the €328/t value of differentiation, or the gap between what green HRC earns as an ordinary commodity and what it earns as a differentiated variety. Without that gap, certified coil is simply expensive steel with no location advantage or productivity margin to carry its fixed cost. Differentiation establishes only that a premium exists. The three models together establish that TKSE can capture it profitably from Australia.

Markusen locates the production. It looks at the three organizational structures to evaluate the highest return profit. In this case, only vertical FDI was profitable, returning €160.1M/year, because producing in Australia lowers the marginal cost of serving the EU by roughly €235/t and avoids the cost of having a German plant entirely. Krugman, then, prices the output. Treated as a differentiated variety, the coil sustains a price of €1,020/t. The resulting markup covers the fixed cost with a break-even quantity of 1.06 Mt/year, well inside the planned scale. Finally, Melitz tests survival. The plant's productivity of 188.2 t/emp/yr clears the survival threshold of 168.8 by a factor of 1.12, with a cost cushion of €59 that widens as CBAM phases in.

Markusen's cost figure becomes Krugman's pricing base and Melitz's productivity bridge. They are not parallel arguments, but rather a single chain, with each model resolving the question the previous one left open. These three models, cumulatively, lead to the decision to build in Western Australia.

3.2 Limitations

The analysis is only as firm as its parameters, and several are estimates rather than measurements, due to the nature of this case study. The most consequential is the elasticity. There is no measured elasticity that I was able to find for green HRC. Therefore, $\varepsilon = 2$ is adopted from Feenstra et al.'s (2018) home versus import macroelasticity rather than estimating from the green-steel market directly. The choice runs into a floor set by willingness to pay. The €310/t premium the differentiated price implies is not a stretch for the buyers this plant is meant to serve. Automakers already sign long-term contracts for certified green coil in the €200–350/t band (Kotas, 2026), and €310/t sits inside it. The broad market is more cautious, and general

buyer surveys still put willingness to pay below that (Fastmarkets, 2024). A premium this size across the whole market waits for the 2030s, once CBAM has lifted the cost of high-carbon imports and Scope 3 reporting gives automotive procurement teams a compliance reason to pay for certified green.

Other parameters, used throughout the study, carry similar caveats. The demand slope, β, is calibrated from a single market equilibrium point rather than estimated econometrically. This means that the demand curve's exact steepness is assumed. Moreover, the freight of €55/t is also an estimate as there is no public rate card for Australia to EU coil. Thirdly, production costs rest on a single project's benchmarks, Port Hedland, rather than a range of operating plants. Finally, the entire study is static, comparing end states at fixed dates rather than modeling the transition path between them.

Two further assumptions are important to highlight. The models treat firms as symmetric. However, in practice, the green-steel field is heterogeneous. Competitors such as SSAB, ArcelorMittal, and Salzgitter all carry different cost structures, timelines, and subsidies. Secondly, the results are sensitive to two prices that have been held fixed. They are the €75/t carbon certificate price, which drives the CBAM schedule, and freight. A lower certificate price or higher freight cost would compress the cost cushion and alter the productivity threshold in the Melitz calculations.

3.3 Implications

Beyond the single firm evaluation of TKSE, the analysis points to three broader concepts. For the market, green steel is emerging as a distinct segment rather than a pricier version of the same commodity. Buyers who need certified low-carbon coil to meet their own Scope 3 targets form a separate demand pool. The firms that enter early earn an outsized margin while supply is still thin. As more producers begin to produce green steel and certify their output, that premium lowers. The advantage, thus, belongs to the early movers.

For the geography of production, the logic that sends TKSE to Western Australia can be expanded beyond just one firm. Integrated steelmaking is starting to move toward places with cheap renewable power and local ore, rather than toward the markets where the steel is sold. Today, Australia, Brazil, and the Middle East are the frontrunner candidates. Further research should ask whether the same logic applies beyond steel, to other energy-intensive goods whose production can be pulled away from their markets and toward cheap, clean power.

For the industry, CBAM is the deciding factor. It does not create TKSE's cost advantage. Instead, it adds rivals' carbon to their landed price on a fixed legal schedule until high-carbon coil becomes the more expensive option in the EU. Consequently, the 2030 phase-in aspect becomes a timing trigger. A plant already operational before that point captures the widening advantage from the beginning, meaning the investment will be worth more to an early entrant. The three models agree that the plant should be built, and CBAM's schedule is what makes building it now, rather than later, the best decision.

Appendix A

Parameters, Sources, and Supporting Data

This appendix documents the value, derivation, and source of every parameter used in the three models. The figures match Table 1 of the main text. The combined fixed cost carried into Krugman and Melitz is $f = G_j + F$ = €539M per year.

Table A1

Shared parameters with derivations and sources

Parameter	Symbol	Value	Derivation and source
Max willingness to pay, EU green HRC	α	€1,040/t	Demand intercept. Fastmarkets (2026b) EU HRC €691/t (Northern Europe index, June 2026) plus green premium €350/t on long-term automotive contracts (Kotas, 2026).
Demand slope	β	€95/t·Mt	Calibrated: $\beta = (\alpha - MC_j)/(2 \cdot q^*) = (1{,}040 - 565)/(2 \times 2.5) = 95$, with $q^* = 2.5$ Mt DR-plant capacity (thyssenkrupp Steel, 2023).
EU market size	L_i	1.0	EU HRC market normalised to 1. Eurofer, European Steel in Figures 2026 (EU HRC ~40 Mt/yr).
Australia market size	L_j	0.1425	Australia 5.7 Mt / 40 Mt EU = 0.1425. IndexBox (June 2025).
German green production cost	MC_i	€800/t	Green H_2-DRI-EAF at Duisburg. Transition Asia et al. (2024), Green Steel Economics – EU (conventional ~€600/t plus green premium ~€180–200/t); Argus (end-2024) opex €910–1,074/t at spot H_2. Central €800/t at H_2 ~€3–4/kg.
Australian production cost	MC_j,local	€510/t	On-site Pilbara ore plus WA solar and wind. Transition Asia et al. (2024), Green Steel Economics – Australia (H_2-DRI-EAF-HSM ~US$516–577/t); CSIRO GenCost 2025–26. Central €510/t (range €500–560).
Australian delivered cost	MC_j	€565/t	Derived: MC_j,local + τ = 510 + 55.
Freight, Australia to EU	τ	€55/t	Finished coil bulk freight plus break-bulk premium. Thunder Said Energy (Capesize ~US$2.5/t per 1,000 nm × ~13,000 nm → US$35–40/t, plus break-bulk and insurance). Central €55/t (range €45–90).
German plant fixed cost	G_i	€329M/yr	€3bn plant, no grant credited, annualised at 7% over 15 years (thyssenkrupp Steel, 2023). Does not enter Type-v.
Australian plant fixed cost	G_j	€439M/yr	€4bn greenfield DRI+EAF+HSM complex ex-power, no subsidy, annualised at 7% over 15 years (CRF 0.11). Benchmarks: Stegra Boden ~€4bn ex-power; ArcelorMittal Gent €1.1bn; Marcegaglia ~€1bn.
Firm-specific fixed cost	F	€100M/yr	G&A overhead for the green line only. 3–5% G&A on ~€2bn line revenue (2.5 Mt × ~€800/t) = €60–100M. Central €100M (thyssenkrupp AG, 2024).
Australian wage	w	€96,000/yr	ABS Average Weekly Earnings, Mining industry (AUD 3,112/wk × 52 ≈ AUD 162,000), converted at 0.593 EUR/AUD [ECB euro reference rate, 2025 average].

Note. All costs are stated in € per ton (€/t) or € million per year (€M/yr).

A.1 Wage and currency

The wage anchors to the ABS Average Weekly Earnings series for the Mining industry, about AUD 3,112 per week in November 2024, or roughly AUD 162,000 per year. Converted at 0.593 EUR/AUD this is about €96,000. The EUR/AUD rate uses the ECB euro reference rate, 2025 average. Labour is a small share of marginal cost, about 5%, so the plant wins on energy and ore rather than on cheap labour. The wage cancels in the Melitz test, because it enters both the plant's productivity and the threshold.

Table A2

Australian wage detail

Item	Value	Source and note
Central wage	€96,000/yr	ABS mining AWOTE, AUD 3,112/wk × 52 ≈ AUD 162,000 at 0.593 EUR/AUD.
Range	AUD ~150,000–165,000/yr	AWOTE AUD 3,112/wk (Nov 2024) rising to AUD 3,174/wk (Nov 2025); FIFO iron-ore roles AUD 100,000–180,000.
Labour share of MC	~4.9%	w ÷ physical output per worker (3,846 t/emp) ≈ €25/t of the €510/t production cost.

Note. The wage is not the source of the plant's advantage. It enters productivity and the threshold together and cancels in the survival comparison (Appendix D).

A.2 Carbon-intensity basis for CBAM

CBAM charges on carbon dioxide only, not on carbon-dioxide-equivalent. The headline intensity for Chinese blast-furnace HRC is 2.7 t CO_2e/t (Hasanbeigi, 2025). About 24% of that total is coal-mining methane, which CBAM does not charge. Stripping it leaves a CBAM-relevant basis of about 2.05 t CO_2/t. This 2.05 figure drives both the Krugman and the Melitz CBAM schedules. TKSE's certified bluemint® coil sits at 0.75 t CO_2/t (DNV-certified).

A.3 CBAM schedule method

The rival add-on in each year is the product of the CBAM-relevant intensity, the certificate price, and the phase-in share. The phase-in share rises on the legislated schedule under Regulation (EU) 2023/956.

$$add\text{-}on = 2.05\ t\ CO_2/t \times €75 \times phase\text{-}in\ share$$

Table A3

CBAM add-on derivation, Chinese blast-furnace HRC

Year	Phase-in share	Rival add-on
2026	2.5%	2.05 × 75 × 0.025 = €3.8/t
2028	10%	2.05 × 75 × 0.10 = €15.4/t
2030	48.5%	2.05 × 75 × 0.485 = €74.6/t
2034	100%	2.05 × 75 × 1.00 = €153.8/t

Note. TKSE's own add-on at 0.75 t CO_2/t is far smaller, about €27/t in 2030 and €56/t in 2034. The same schedule enters both model sections.

A.4 Supporting data series

The two series below are context for the cost figures and are not used directly in the calculations. Table A4 explains why the German benchmark MC_i is elevated. Table A5 gives the hydrogen trajectory behind the falling Australian marginal cost in the Melitz time-dynamics table (Appendix D).

Table A4

German industrial electricity price history

Year	German industrial electricity	Note
2018	~15–16 ct/kWh	Pre-crisis baseline
2021	21.38 ct/kWh	Pre-war level, already elevated
2022	29.01 ct/kWh	Peak; loss of Russian gas
2024	23.3 ct/kWh	About 25% above the EU average (18.7 ct/kWh)
2025	11.69 ct/kWh	Partial recovery, eligible large consumers only

Note. Sources: Bundesnetzagentur SMARD, 2026 (Feb 2025 data); Eurostat and Cube Concepts (Apr 2026); Open Energy Tracker (2026). Elevated German power underlies the €800/t green H_2-DRI-EAF benchmark for MC_i.

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Table A5

Australian green-hydrogen cost trajectory

Year	Green H_2 cost (AUD/kg)	Effect on $MC_{j,local}$	Source
2025	5.70–6.10	Base case	PwC Australia (2023)
2030	4.10–4.50	~$88/t reduction	PwC Australia; IEA H_2 projections
2040	2.00–2.45	~$200/t reduction vs 2025	PwC Australia; CSIRO GenCost 2025–26

Note. At about 55 kg H_2 per ton of DRI, each AUD 1/kg fall in hydrogen lowers $MC_{j,local}$ by about AUD 55/t. This trajectory drives the falling marginal cost in Table D2.

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Appendix B

Markusen Model: Profit Calculations

This appendix shows the full working behind Table 2 of the main text. The profit of a linear-demand monopolist in each market m is $\Pi_m = (\alpha - MC_m)^2 \cdot L_m/(4\beta)$, with $4\beta = 4 \times 95 = 380$. Total profit sums the two market segments net of the fixed costs of each structure. Three structures are compared.

B.1 Type-d: produce in Germany, export to Australia

EU segment, no freight: $(1{,}040 - 800)^2 \times 1.0 / 380 = 240^2 / 380 = 57{,}600 / 380 = €151.6M$.

Australia segment, freight τ added: $(1{,}040 - 800 - 55)^2 \times 0.1425 / 380 = 185^2 \times 0.1425 / 380 = 34{,}225 \times 0.1425 / 380 = €12.8M$.

Fixed costs: $G_i + F = 329 + 100 = 429$. Total: $\Pi^d = 151.6 + 12.8 - 429 = -€264.6M/yr$.

B.2 Type-h: plants in both countries

EU segment, Germany at MC_i: €151.6M, as in Type-d.

Australia segment, local production at MC_j,local with no freight: $(1{,}040 - 510)^2 \times 0.1425 / 380 = 530^2 \times 0.1425 / 380 = 280{,}900 \times 0.1425 / 380 = €105.3M$.

Fixed costs: $G_i + G_j + F = 329 + 439 + 100 = 868$. Total: $\Pi^h = 151.6 + 105.3 - 868 = -€611.1M/yr$.

B.3 Type-v: produce in Australia, ship finished coil to EU

EU segment, delivered at MC_j = €565: $(1{,}040 - 565)^2 \times 1.0 / 380 = 475^2 / 380 = 225{,}625 / 380 = €593.8M$.

Australia local segment at MC_j,local = €510: €105.3M, as in Type-h.

Fixed costs: $G_j + F = 439 + 100 = 539$. Total: $\Pi^v = 593.8 + 105.3 - 539 = +€160.1M/yr$.

Table B1

Arithmetic spot checks

Expression	Value
4×95	380
240^2	57,600
$185^2 \cdot \times 0.1425 \cdot \div 380$	$34{,}225 \cdot 4{,}877 \cdot 12.8$
475^2	225,625
$530^2 \cdot \times 0.1425 \cdot \div 380$	$280{,}900 \cdot 40{,}028 \cdot 105.3$

Note. All segment contributions are stated in €M/yr before fixed costs. Type-v is the only profitable structure.

B.4 Robustness

Holding all other parameters fixed, Type-v stays the most profitable structure until the Australian plant fixed cost G_j reaches about €599M per year, which corresponds to roughly €5.5bn of capex, far above the €439M assumed. Fixed costs are built bottom-up from sourced capex and annualised at 7% over 15 years, with no subsidies credited to either plant. Crediting the larger subsidies available to a German plant would raise all profit levels but narrow the Type-v lead, since the German plant is more heavily subsidised. The ranking does not change either way.

Appendix C

Krugman Model: Calculations

This appendix sets out the course formulas, the value-of-differentiation calculation, the full sensitivity band, and the marginal-cost robustness check. The elasticity of substitution ε is the only free parameter. The base case is ε = 2.0 (Feenstra et al., 2018), with a 1.8–2.5 band.

C.1 Course formulas

Table C1

Krugman formulas in course notation

Object	Formula
Markup price	$p^*_i = (\varepsilon/(\varepsilon-1))\cdot(w/\varphi_i) = (\varepsilon/(\varepsilon-1))\cdot MC$
Zero-profit quantity	$q^*_i = (\varepsilon-1)\cdot f / MC$
Number of varieties	$n^* = wL / (\varepsilon\cdot f)$
CES demand for variety i	$Demand_i = (wL/P)\cdot(p_i/P)^{-e}$
CES price index	$P = [\ \Sigma_j\ p_j^{(1-e)}\ ]^{(1/(1-\varepsilon))}$

Note. $f = G_j + F$ = €539M/yr. wL ≈ €25bn, the EU HRC market. The superscript on the demand and price-index terms is the elasticity ε.

C.2 Value of differentiation

The markup rule prices the two elasticity regimes at the same marginal cost of €510/t.

Commodity regime, ε = 3.8: markup = 3.8/2.8 = 1.357; p* = 1.357 × 510 = €692/t. This sits €18 below the €710/t standard-HRC benchmark, so there is no premium.

Differentiated regime, ε = 2.0: markup = 2.0/1.0 = 2.000; p* = 2.000 × 510 = €1,020/t, a €310 premium over €710.

The gap between the two regime prices, €1,020 − €692 = €328/t, is the economic value of differentiation.

C.3 Base case and sensitivity band

Worked example at ε = 2.0: q* = (2.0 − 1) × 539 / 510 = 539 / 510 = 1.06 Mt. n* = 25,000 / (2.0 × 539) = 25,000 / 1,078 = 23.2. Internal check: n*·p*·q* ≈ 23 × €1,020 × 1.06 Mt ≈ €25bn = wL.

Table C2

Base case and sensitivity band, differentiated regime, f = €539M

ε	markup	p*	premium vs €710	q* (Mt)	n*	note
2.5	1.667	€850	+€140	1.59	18.6	conservative, partial differentiation
2.3	1.769	€902	+€192	1.37	20.2	mid-band
2.0	2.000	€1,020	+€310	1.06	23.2	base case (Feenstra macro; WTP floor)
1.8	2.250	€1,148	+€438	0.85	25.8	below WTP floor, excluded

Note. q* from $q^* = (\varepsilon-1)\cdot f/MC$ and n* from $n^* = wL/(\varepsilon\cdot f)$, at f = €539M, MC = €510/t, wL ≈ €25bn. The commodity anchor ε = 3.8 gives p* = €692 and q* = 2.96 Mt, below the €710 benchmark. ε = 1.8 is reported only to mark the lower bound.

C.4 Marginal-cost robustness

The base case uses the Australian production cost of €510/t, the marginal cost shared with Markusen and Melitz. Using the EU-delivered cost of €565/t instead raises the markup base. At €565, p* rises to €1,130/t, q* falls to 0.95 Mt, and the Melitz cost cushion compresses to about €3/t. The €510 base is retained and the €565 figure is reported as the sensitivity.

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Appendix D

Melitz Model: Calculations

This appendix derives the break-even cost and the survival threshold in full, states the value of the second market, and gives the time-dynamics table behind Table 6 of the main text.

D.1 Break-even cost

The winning Type-v structure survives while $\Pi^v \geq 0$. Setting $\Pi^v = 0$ and writing the Australian production cost as c, with EU delivery at c + τ, the profit function becomes a quadratic in u = α − c.

$$u^2(L_i + L_j) - 2u \cdot \tau \cdot L_i + [\, \tau^2 \cdot L_i - 4\beta(G_j + F) \,] = 0$$

Coefficients: $A = L_i + L_j = 1.1425$. $B = -2 \cdot \tau \cdot L_i = -2 \times 55 \times 1.0 = -110$. $C = \tau^2 \cdot L_i - 4\beta(G_j + F) = 55^2 \times 1 - 380 \times 539 = 3{,}025 - 204{,}820 = -201{,}795$.

Discriminant: $B^2 - 4AC = 12{,}100 - 4(1.1425)(-201{,}795) = 12{,}100 + 922{,}203 = 934{,}303$; $\sqrt{} = 966.59$.

Admissible root: u = (110 + 966.59) / (2 × 1.1425) = 1,076.59 / 2.285 = 471.16. The other root is negative and describes a plant that could never price its output at a profit, so it is discarded.

Break-even production cost: c* = α − u = 1,040 − 471.16 = €568.84/t. The delivered break-even cost is c* + τ = €623.84/t.

D.2 Verification

Substituting c* back into the profit function returns zero. EU segment at delivered €623.84: $(1{,}040 - 623.84)^2 / 380 = 416.16^2 / 380 =$ €455.8M. Australia segment at €568.84: $(1{,}040 - 568.84)^2 \times 0.1425 / 380 = 471.16^2 \times 0.1425 / 380 =$ €83.2M. Total: $\Pi^v = 455.8 + 83.2 - 439 - 100 =$ €0.0M. The solution checks.

D.3 Threshold productivity

Melitz re-expresses the cost ceiling as a productivity floor through MC = w/φ, so φ* = w/c*.

$$\varphi^* = w / c^* = 96{,}000 / 568.84 = 168.8 \text{ t/emp/yr}$$

The plant's model-consistent productivity is φ = w/MC = 96,000 / 510 = 188.2 t/emp/yr. The plant clears the threshold, 188.2 against 168.8, a factor of 1.12. In cost terms, €510/t sits €58.84/t below the €568.84/t ceiling, about a 10% cushion.

Table D1

Melitz survival test in productivity and cost forms

Measure	Value	vs threshold	Verdict
Model productivity φ = w/MC	188.2 t/emp/yr	1.12× over	clears the bar
Threshold productivity φ*	168.8 t/emp/yr	—	minimum to survive
TKSE Australian cost (MC)	€510/t	€58.84/t under	clears on cost
Threshold cost c*	€568.84/t	—	maximum cost to survive

Note. φ = w/MC and φ* = w/c* at w = €96,000/yr. The productivity and cost rows are the same test in two units. The physical benchmark below is descriptive only and is not used in the test (see D.5).

D.4 Value of the second market

If the plant had to cover all fixed cost from EU sales alone, the single-market break-even would be $c^*_{single} = \alpha - \tau - \sqrt{[4\beta(G_j + F)/L_i]}$.

Working: $4\beta(G_j + F)/L_i$ = 380 × 539 / 1.0 = 204,820; √ = 452.57. So c*single = 1,040 − 55 − 452.57 = €532.43/t, and φ*single = 96,000 / 532.43 = 180.3 t/emp/yr.

Serving the Australian domestic market lowers the required productivity by 11.5 t/emp/yr, from 180.3 to 168.8. Australia is import-dependent for flat steel and has no domestic green-HRC producer, so local sales earn full margin and absorb part of the fixed cost. This is the Melitz entry-with-export logic.

D.5 Physical benchmark, descriptive only

The plant's physical output per worker is about 3,846 t/emp/yr, or 2.5 Mt from roughly 650 staff, benchmarked from the POSCO Port Hedland project. This describes how automated a modern DRI+EAF+HSM line is. It is not comparable to φ*. The threshold φ* = w/c* treats labour as the whole of marginal cost, as the single-input cost function requires, whereas in the physical plant labour is only about 5% of marginal cost. The survival test therefore rests on the cost comparison, €510 against €568.84, and its model-consistent form, 188.2 against 168.8, not on the physical figure.

D.6 Time dynamics

Three forces raise the break-even ceiling c* and lower the required φ* over 2026 to 2034. Green hydrogen costs fall, which lowers the Australian marginal cost. CBAM raises rivals' effective EU prices, which lifts the effective demand intercept α_eff. Operational learning and bluemint® customer relationships lower the plant's effective cost. The cost cushion below break-even widens from about €59/t in 2026 to €240/t by 2034, about 4.1 times.

Table D2

Cost cushion widening through 2034 (ε held constant)

Year	MC (€/t)	α_eff (€/t)	c* (€/t)	φ* (t/emp/yr)	Cushion (c*−MC)
2026 (base)	510	1,040	568.8	168.8	€59/t (—)
2028	480	1,045	573.8	167.3	€94/t (1.6×)
2030 (CBAM phase-in)	430	1,065	593.8	161.7	€164/t (2.8×)
2034	380	1,091	619.8	154.9	€240/t (4.1×)

Note. c* = α_eff − 471.16, the break-even gap from D.1; φ* = w/c* at w = €96,000/yr. α_eff rises by about one-third of the rival CBAM add-on (Table A3); MC follows the PwC Australia green-H_2 trajectory (Table A5). Cushion = c* − MC.

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References

Ahmad, S., Montgomery, C., & Schreiber, S. (2021). A comparison of sectoral Armington elasticity estimates in the trade literature. *Journal of International Commerce and Economics*. U.S. International Trade Commission. https://www.usitc.gov/publications/332/journals/jice_armington_elasticity.pdf

Argus Media. (2024). *Green hydrogen-based DRI-EAF steelmaking cost assessment* [Report]. https://www.argusmedia.com

Australian Bureau of Statistics. (2025). *Average weekly earnings, Australia, November 2024*. https://www.abs.gov.au/statistics/labour/earnings-and-working-conditions/average-weekly-earnings-australia/nov-2024

Bajzik, J., Havranek, T., Irsova, Z., & Schwarz, J. (2020). Estimating the Armington elasticity: The importance of study design and publication bias. *Journal of International Economics, 127*, 103383. https://doi.org/10.1016/j.jinteco.2020.103383

Broda, C., & Weinstein, D. E. (2006). Globalization and the gains from variety. *The Quarterly Journal of Economics, 121*(2), 541–585. https://doi.org/10.1162/qjec.2006.121.2.541

Bundesnetzagentur. (2026). *Electricity market data* [SMARD]. https://www.smard.de

CBAM Guide. (2026). *CBAM certificate price tracker: Official certificate price and EU ETS reference*. https://cbamguide.com/tools/price-tracker/

Commonwealth Scientific and Industrial Research Organisation. (2025). *GenCost 2025–26 final report*. CSIRO & Australian Energy Market Operator. https://www.csiro.au/en/research/technology-space/energy/Electricity- transition/GenCost

Cube Concepts. (2026). *Europe's industrial electricity prices 2024*. https://cubeconcepts.de/en/europes-industrial-electricity-prices-2024/

Department of Industry, Science and Resources. (2025). *New fund will position Australia at the centre of the global green iron market*. Australian Government. https://www.industry.gov.au/news/new-fund-will-position-australia-centre-global-green-iron-market

Devlin, A., Kossen, J., Goldie-Jones, H., & Yang, A. (2023). Global green hydrogen-based steel opportunities surrounding high quality renewable energy and iron ore deposits. *Nature Communications, 14*, 2578. https://doi.org/10.1038/s41467-023-38123-2

Discovery Alert. (2026). *European green steel demand surge driven by regulatory changes*. https://discoveryalert.com.au/green-steel-europe-demand-market-2026/

European Central Bank. (2025). *Euro foreign exchange reference rates: Australian dollar (AUD)* [Data set; 2025 average]. https://www.ecb.europa.eu/stats/policy_and_exchange_rates/euro_reference_exchange_rates/html/eurofxref-graph-aud.en.html

European Steel Association (Eurofer). (2026). *European steel in figures 2026*. https://www.eurofer.eu/publications/brochures-booklets-and-factsheets/european-steel-in-figures-2026

Eurostat. (2026). *Electricity prices for non-household consumers* [nrg_pc_205] [Data set]. https://ec.europa.eu/eurostat/databrowser/view/nrg_pc_205/default/table

Fastmarkets. (2024, September 5). *Green steel premiums not enough to offset technology costs in the short term: China Steel Congress*. https://www.fastmarkets.com/insights/green-steel-premiums-not-enough-to-offset-technology-costs-in-the-short-term/

Fastmarkets. (2026a). *Global green steel markets in 2026: Regulation, costs and regional divergence*. https://www.fastmarkets.com/insights/global-green-steel-markets-in-2026-regulation-costs-and-regional-divergence/

Fastmarkets. (2026b). *Steel hot-rolled coil index, domestic, exw Northern Europe (MB-STE-0028)* [Price assessment]. Fastmarkets. https://www.fastmarkets.com/

Fastmarkets. (2026c). *Why automakers are set to lead green steel adoption in Europe*. https://www.fastmarkets.com/insights/why-automakers-are-set-to-lead-green-steel-adoption-in-europe/

Feenstra, R. C., Luck, P., Obstfeld, M., & Russ, K. N. (2018). In search of the Armington elasticity. *The Review of Economics and Statistics, 100*(1), 135–150. https://doi.org/10.1162/REST_a_00696

Geoscience Australia. (2025). *Australia's identified mineral resources 2025*. Australian Government. https://www.ga.gov.au/aimr2025

GMK Center. (2026, April 9). *Western Australia has approved POSCO's plan to build an HBI plant*. https://gmk.center/en/news/western-australia-has-approved-posco-s-plan-to-build-an-hbi-plant/

Hasanbeigi, A. (2025). *Steel climate impact 2025: An international benchmarking of energy and GHG intensities*. Global Efficiency Intelligence. https://www.globalefficiencyintel.com/steel-climate-impact-2025-an-international-benchmarking-of-energy-and-ghg-intensities

IndexBox. (2025). *Australia – flat-rolled steel in coils: Market analysis, forecast, size, trends and insights*. https://www.indexbox.io/store/australia-and-oceania-flat-hot-rolled-steel-in-coils-market-analysis-forecast-size-trends-and-insights/

International Energy Agency. (2024). *Global hydrogen review 2024*. IEA. https://www.iea.org/reports/global-hydrogen-review-2024

Johnson, C., Åhman, M., Nilsson, L. J., & Li, Z. (2025). Emerging green steel markets surrounding the EU emissions trading system and carbon border adjustment mechanism. *Nature Communications, 16*, 9087. https://doi.org/10.1038/s41467-025-64440-9

Kotas, A. M. (2026). *Green steel premiums: Promise or illusion?* Steelonthenet. https://doi.org/10.5281/zenodo.20526611

Krugman, P. R. (1980). Scale economies, product differentiation, and the pattern of trade. *The American Economic Review, 70*(5), 950–959.

Markusen, J. R. (2002). *Multinational firms and the theory of international trade*. MIT Press.

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Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica, 71*(6), 1695–1725. https://doi.org/10.1111/1468-0262.00467

Open Energy Tracker. (2026). *Germany: Energy prices*. https://openenergytracker.org/en/docs/germany/prices/

PwC Australia. (2023). *Getting H2 right: Success factors for Australia's hydrogen export industry*. https://www.pwc.com.au/energy-transition/getting-h2-right-australias-competitive-hydrogen-export-industry.html

Regulation (EU) 2023/956 of the European Parliament and of the Council of 10 May 2023 establishing a carbon border adjustment mechanism, OJ L 130, 16.5.2023, pp. 52–104. http://data.europa.eu/eli/reg/2023/956/oj

Thunder Said Energy. (2023). *Bulk shipping cost breakdown*. https://thundersaidenergy.com/downloads/bulk-shipping-cost-breakdown/

thyssenkrupp AG. (2024). *Annual report 2023/2024*. https://d2zo35mdb530wx.cloudfront.net/_binary/thyssenkruppAGReport/354e8b26-9f91-40f1-80cf-91207a65c9d0/Annual-report_2023-2024.pdf

thyssenkrupp Steel. (2023, March 1). *thyssenkrupp Steel awards a contract worth billions of euros to SMS group for a direct reduction plant: One of the world's largest industrial decarbonization projects gets underway* [Press release]. https://www.thyssenkrupp-steel.com/en/newsroom/press-releases/thyssenkrupp-steel-awards-a-contract-worth-billions-of-euros-to-sms-group-for-a-direct-reduction-plant.html

Transition Asia, Global Efficiency Intelligence, & Solutions for Our Climate. (2024). *Green steel economics: Comparing economics of green H2-DRI and traditional steelmaking around the world*. Transition Asia. https://transitionasia.org/green-steel-economics/

U.S. Geological Survey. (2025). Iron ore. In *Mineral commodity summaries 2025*. U.S. Geological Survey. https://doi.org/10.3133/mcs2025

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